Name: _____

WORKSHEET

Dartboard probabilities

Consider a standard dartboard. It has dimensions: outside diameter = 450 mm, scoring area diameter = 340 mm, inner bull diameter = 12.7 mm, outer bull diameter = 31.8 mm, outer treble ring diameter = 214 mm, double/treble ring width = 9 mm.

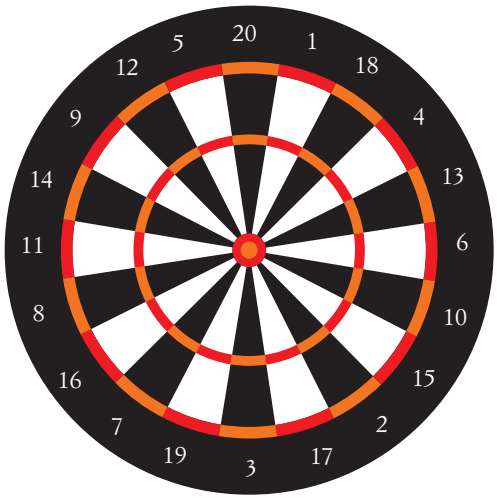


Diagram not to scale

1 If a player randomly throws a dart at the board, they have an equal chance of hitting anywhere on the entire dartboard. Assuming they hit the board find:

a P(scoring)

b P(inner bullseye)

c P(20 or double 20 or triple 20)

d P(outer or inner bullseye)

e P(outer bullseye only)

f P(within the outer treble ring)

2 A skilled player aims for the centre of the inner bullseye, and has an equal chance of hitting anywhere within a given radius of where they aim, depending on their skill level.

a Find the probability that they hit the inner bullseye if the dart lands within a radius of:

i 5 mm

ii 9 mm

iii 25 mm

iv 50 mm

© Cengage Learning Australia Pty Ltd 2013

b Find the probability that they hit the outer bullseye if the dart lands within a radius of:

i 9 mm

ii 25 mm

iii 50 mm

iv 80 mm

3 For the players below find the radius in which their dart lands if their probability of aiming for and hitting the inner bullseye is

a 0.02 (beginner)

b 0.2 (social player)

c 0.5 (competitor)

d 0.9 (professional)

© Cengage Learning Australia Pty Ltd 2013

Answers

1 **a** 0.57086
b 0.000796
c 0.02829
d 0.00499
e 0.004197
f 0.22615

2 **a** **i** 1
ii 0.4978
iii 0.0645
iv 0.0161
b **i** 0.50219
ii 0.33998
iii 0.084
iv 0.03320

3 **a** $r = 45$ mm
b $r = 14.2$ mm
c $r = 9$ mm
d $r = 6.7$ mm

© Cengage Learning Australia Pty Ltd 2013